Minsung Kim

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Education

University of Toronto - Computer Science Specialist September 2025 – June 2029
GPA 3.7/4 | *University of Toronto International Scholar Award*

Skills

Languages: C/C++, C#, Python, Java, TypeScript, JavaScript, SQL, HTML5/CSS, PostgreSQL, MySQL, C/C++, C#, Bash, R
Tools/Frameworks: React, Next.js, Vite, Zustand, Tailwind CSS, Material UI, jQuery, Express.js, RESTful APIs, WebSockets, PostgreSQL, MySQL, Supabase, Python, Scikit-learn, K-Means, Open CV, AWS, Git, GitHub, Linux, Postman, Bcrypt

Project

UofT Student Life Tracker July 2026 – August 2026
• Developed a student dashboard integrating UofT resources using React, Express, PostgreSQL, and Material UI to centralize the navigation between separate platforms
Engineered a RESTful API for calendar and deadline management, reducing academic planning time by approximately 40% through centralizing event, task, and deadline CRUD workflows into a unified web platform
• Secured user authentication by implementing server-side Express sessions, hardened connect.sid cookies, and bcrypt password hashing

Cluster Grade AI March 2026
Hack the Student Life *AWS Services & University of Toronto*
• Developed Cluster Grade AI with a team of three, an AI grading platform that extracts text from scanned handwritten assignments using AWS, Python Scikit-Learn, and Sentence Transformers to automate analysis of student submissions
• Built an OCR-to-clustering pipeline that converts scanned assignments into semantic embeddings and applies K-Means to group recurring logic errors, achieving a 0.72 silhouette score across 10 test submissions
• Integrated Amazon Bedrock with instructor rubrics to generate rubric-aligned deductions and feedback, allowing one TA review to serve entire groups instead of rewriting repetitive feedback
• Designed a human-in-the-loop grading workflow requiring TA approval before applying AI-generated feedback, targeting a 40–60% reduction in grading time for large introductory courses

WatchLine March 2026
Hack Canada *SPUR Innovation*
• Collaborated in a team of three to create a real-time AI transit safety system using YOLOv8-Pose, Python, and open CV to track human pose and classify falls, aggression, crouching, lying down, and erratic movement from live camera streams
• Built an asynchronous FastAPI and WebSocket backend that processes live video to maintain per-person motion histories, records critical incidents, and broadcasts the video clips to connected clients in real time
• Integrated Google Gemini 2.0 Flash to convert video evidence of detected incidents into English dispatch summaries, storing data with Supabase/PostgreSQL
• Developed a monitoring dashboard with React, Vite, and Zustand that renders MJPEG camera streams, displays real-time safety alerts, and provides incident clip playback for rapid event review

Experience

ResNet Technical Assistant May 2026 – Present
University of Toronto
• Troubleshoot Wi-Fi, Ethernet, and eduroam connectivity issues across Chestnut, Oak House, and Graduate House, resolving network access problems for residential users
• Assist residents with wired and wireless device connectivity, including Ethernet cables, USB-A/USB-C adapters, HDMI cables, and gaming consoles requiring MAC address registration
• Monitor residential network infrastructure through regular access-point and connectivity checks, while restoring disconnected Ethernet and university phone lines through troubleshooting and power cycling
• Troubleshooted eduroam authentication issues on Android devices, configuring CA certificates and network security settings to restore connectivity, while providing guest-network alternatives when device configurations prevented authentication.